

Goal Derivation

How a client's Protection / Income / Growth goal is inferred from the balance sheet — methodology, formula, and a worked example for every client

Prepared	Scope	Source of truth	Status
18 June 2026	deriveGoals() — inferred goal vector	goals.js · data.js	Live engine · additive (not yet wired into fit scoring)

The one-line idea. The old model assumed the client *handed you* a target split. Real clients don't. So `GOALS.deriveGoals(client)` **infers** a three-bucket goal — **Protection · Income · Growth** — from **both sides of the balance sheet** plus the client's **revealed risk appetite**, never from what they own for its own sake. Every number below was produced by running the actual engine (`goals.js` + `PARAMS`) against the real seed book. They are exact, not illustrative.

1 · The reframe — “have” vs “need”

The engine separates two things the old `goals.target` conflated:

- **Current allocation (“have”)** — what the book holds today. Read straight off positions.
- **Goal allocation (“need”)** — what the book *should* hold, given the client's obligations, horizon and appetite. This is what `deriveGoals` computes — and what replaces the hand-typed target.

The methodology is **liability-driven / goals-based** (after Chhabra's risk-allocation framework, where the Safety / Market / Aspirational buckets map onto **Protection / Income / Growth**). The principle: **fund the obligations first, then deploy the surplus by how hard the money must work.**

1.1 The evidence ladder — the `source` field

The inference uses the strongest evidence available and records which tier it leaned on, in priority order:

source	Has a \$ goal?	Has a risk descriptor?	What it leans on	Confidence
STATED- GOAL	✓ → required return	usually	The funding number (objective — how hard the money must work) + willingness	0.90
STATED- RISK	✗	✓ → willingness	The written appetite (“aggressive” / “conservative”) blended with the	0.70

			book	
REVEALED	X	X	The portfolio alone — there is no form to read off	0.55

A **goal** is a number (“grow to \$200m by 2045”) → it gives a *required return*. A **risk descriptor** is temperament in words (“aggressive”) → it gives *willingness*, but says nothing about the target. **stated-risk** = appetite known, goal unknown.

2 • The formula

Each of the three buckets gets a raw score built from measured drivers; the three are then normalised to 100%. Every *input* is a real quantity off the balance sheet; every *coefficient* lives in `PARAMS` and is a calibration choice.

2.1 Willingness — read two ways, then blended

Willingness (0–1) is the risk-appetite knob. It is read from the current book (**revealed**) and, if present, from the free-text risk descriptor (**stated**):

```
Revealed = 0.5·R + 0.3·D + 0.2·C
  R = risk-asset share      (Equity + Alternatives, as a fraction of the book)
  D = low-defensiveness    = 1 - min(1, defensiveShare / 0.55)  defensive = Cash +
Fixed Income + Commodity
  C = concentration appetite = min(1, topSingleName / 0.25)      (index cores /
fundless sleeves excluded)

Stated = aggressive→0.95 · growth→0.80 · moderate→0.50 · conservative/income→0.30 ·
(none)→null
Willingness = (Stated is null) ? Revealed : 0.5·Stated + 0.5·Revealed
```

2.2 The three raw scores

```
caution = 1 - Willingness      // willingness is BIDIRECTIONAL: low appetite adds
ballast

PROTECTION = 8 + 1.0·nearBill% + 0.3·debtLoad% + max(0, topName%-15)
              + 15·[drawdown] + 3·max(0, 5-H) + 28·caution

INCOME      = 8 + 8·incomeNeed% + 8·service% + 0.6·debtLoad%
              + 15·[drawdown] + 18·caution

GROWTH      = 8 + 5·requiredReturn% + 3·max(0, H-6) + Wv·Willingness
              Wv = 30 if a funding goal exists, else 45 (lean harder on appetite when
no goal)

vector = normalise( Protection, Income, Growth ) → sums to 100
```

Symbol	Meaning (all measured off the client)	Drives
nearBill%	One-off bills due soon (tax, bridge), % of AUM	Protection
debtLoad%	Total liabilities, % of AUM	Income (match) + Protection (ballast)
service%	Annual debt servicing (balance × ~6%), % of AUM	Income

incomeNeed%	Stated annual spending draw, % of AUM	Income
topName%	Largest single-name position (index cores excluded)	Protection (above 15%)
H	Horizon midpoint, years; [drawdown] = spend-down phase	Growth (long) / Protection (short, drawdown)
requiredReturn%	CAGR needed to hit the funding goal: $(\text{target}/\text{base})^{(1/\text{yrs})} - 1$	Growth

Why willingness is bidirectional. Originally willingness only fed Growth, so a cautious client with no liabilities or income needs still came out growth-heavy (Protection/Income only ever got weight from *needs*). The `caution = 1 - willingness` term routes a *low* appetite into ballast: full-aggressive pushes ~45 points into Growth, full-cautious pushes ~46 (28+18) into Protection+Income — a balanced two-way dial.

2.3 Anti-circularity

The book sets the willingness *knob* (a scalar), **not** the target vector. The vector is always rebuilt by the formula, so the gap between “have” and “need” stays meaningful — the engine never trivially concludes “you already own the right thing.” A manual `client.goalOverride` pins a vector and short-circuits the model.

3 • Worked examples — every client

Each block shows the measured inputs, the willingness build, the three raw scores term-by-term, and the normalised goal. Numbers are exact from `goals.js`.

Fable STATED-GOAL • Professional • \$38.0m • “Aggressive” • horizon 10y

Funding: “Grow to \$75m by 2034”. AI-concentrated book — 88% equity, a 24% single name.

$$\text{Willingness} = 0.5 \cdot 0.95 \text{ (stated)} + 0.5 \cdot [0.5 \cdot 0.88(R) + 0.3 \cdot 0.891(D) + 0.2 \cdot 0.96(C)] = 0.899] = \mathbf{0.925}$$

```
requiredReturn = (75/38)^(1/8) - 1 = 8.87%/yr
PROTECTION = 8 + 9.0 (24%-15 conc) + 2.1 (caution 28·0.075)
= 19.1
INCOME      = 8 + 1.35 (caution 18·0.075)
= 9.4
GROWTH      = 8 + 44.35 (5·8.87) + 12 (3·[10-6]) + 27.75 (30·0.925)
= 92.1
→ normalise(19.1, 9.4, 92.1)
```

→ **Protection 16 • Income 8 • Growth 76**

High required return (8.9%) and high appetite drive Growth; the 24% single name is the only real pull toward Protection.

Aurora STATED-GOAL • Retail • €50.2m • “Growth, with income needs” • horizon 8.5y

Funding: “Grow to €70m by 2035 while drawing income”. Concentrated 25.8% Micron.

$$\text{Willingness} = 0.5 \cdot 0.80 \text{ (stated)} + 0.5 \cdot [0.5 \cdot 0.709(R) + 0.3 \cdot 0.471(D) + 0.2 \cdot 1.0(C)] = 0.696] = \mathbf{0.748}$$

```
requiredReturn = (70/50.2)^(1/9) - 1 = 3.76%/yr
PROTECTION = 8 + 10.8 (25.8%-15 conc) + 7.06 (caution 28·0.252)
= 25.9
INCOME      = 8 + 4.54 (caution 18·0.252)
= 12.5
GROWTH      = 8 + 18.8 (5·3.76) + 7.5 (3·[8.5-6]) + 22.44 (30·0.748)
= 56.8
→ normalise(25.9, 12.5, 56.8)
```

→ **Protection 27 • Income 13 • Growth 60**

Known limitation: her “income needs” is qualitative with no \$ figure, so Income reads only 13. A spending field would lift it. The 25.8% Micron lump drives Protection to 27.

Scott **STATED-RISK** · Retail · \$124.0m · “Conservative income” · drawdown, horizon 2.5y

Funding is income-only (“Fund \$3.6m/yr”) → no required-return signal. Retiree, spending down.

Willingness = $0.5 \cdot 0.30$ (stated) + $0.5 \cdot [0.5 \cdot 0.30(R) + 0.3 \cdot 0(D) + 0.2 \cdot 0.36(C) = 0.222]$
= **0.261**

```
incomeNeed = 3.6/124 = 2.9% of book · drawdown phase ·  $W_v = 45$  (no goal)
PROTECTION = 8 + 15 (drawdown) + 7.5 (3·[5-2.5] short H) + 20.69 (caution 28·0.739) = 51.2
INCOME      = 8 + 23.2 (8·2.9 draw) + 15 (drawdown) + 13.3 (caution 18·0.739) = 59.5
GROWTH      = 8 + 11.75 (45·0.261) = 19.7
→ normalise(51.2, 59.5, 19.7)
```

→ **Protection 39 · Income 46 · Growth 15**

Textbook preservation book: a real income draw, drawdown phase and low appetite all pile into Income and Protection. Growth collapses to 15.

Amar **STATED-GOAL** · Professional · \$71.0m · “Growth, real-asset tilt” · horizon 10y

Funding: “Preserve real value, grow to \$90m by 2036”. Real-asset / commodity-heavy, BTC sleeve.

Willingness = $0.5 \cdot 0.80$ (stated) + $0.5 \cdot [0.5 \cdot 0.49(R) + 0.3 \cdot 0.418(D) + 0.2 \cdot 0.64(C) = 0.498]$ = **0.649**

```
requiredReturn = (90/71)^(1/10) - 1 = 2.40%/yr
PROTECTION = 8 + 1.0 (16%-15 conc) + 9.83 (caution 28·0.351) = 18.8
INCOME      = 8 + 6.32 (caution 18·0.351) = 14.3
GROWTH      = 8 + 12.0 (5·2.4) + 12 (3·[10-6]) + 19.47 (30·0.649) = 51.5
→ normalise(18.8, 14.3, 51.5)
```

→ **Protection 22 · Income 17 · Growth 61**

A modest required return (2.4%) is offset by a long horizon and a growth appetite; the diversified book keeps Protection/Income moderate.

Jacob **STATED-GOAL** · Professional · \$96.0m · “Moderate” · horizon 9y · liabilities

Funding: “Net worth to \$120m by 2033”. \$19m mortgage + \$2.4m Q3 tax — the balance-sheet client.

Willingness = $0.5 \cdot 0.50$ (stated) + $0.5 \cdot [0.5 \cdot 0.58(R) + 0.3 \cdot 0.509(D) + 0.2 \cdot 0.40(C)] = 0.523$ = **0.511**

```
requiredReturn = (120/75)^(1/7) - 1 = 6.94%    · debtLoad 22.3%    · nearBill 2.5%
· service 1.19%
PROTECTION = 8 + 2.5 (nearBill tax) + 6.69 (0.3·22.3 debt ballast) + 13.69
(caution 28·0.489)    = 30.9
INCOME      = 8 + 9.52 (8·1.19 service) + 13.37 (0.6·22.3 debt match) + 8.8
(caution 18·0.489)    = 39.7
GROWTH      = 8 + 34.7 (5·6.94) + 9 (3·[9-6]) + 15.33 (30·0.511)
= 67.1
→ normalise(30.9, 39.7, 67.1)
```

→ **Protection 22 · Income 29 · Growth 49**

The \$19m floating mortgage does real work — ~20 points into Income (service + matching) and Protection (ballast + the tax reserve) — pulling a high-required-return book back to balanced.

Prahnav **STATED-GOAL** · Retail · \$44.0m · “Growth” · horizon 8.5y

Funding: “Grow to \$70m by 2034; \$4m house deposit by 2028”. Concentrated 22% NVDA.

Willingness = $0.5 \cdot 0.80$ (stated) + $0.5 \cdot [0.5 \cdot 0.70(R) + 0.3 \cdot 0.60(D) + 0.2 \cdot 0.88(C)] = 0.706$ = **0.753**

```
requiredReturn = (70/44)^(1/8) - 1 = 5.98%/yr
PROTECTION = 8 + 7.0 (22%-15 conc) + 6.92 (caution 28·0.247)
= 21.9
INCOME      = 8 + 4.45 (caution 18·0.247)
= 12.4
GROWTH      = 8 + 29.9 (5·5.98) + 7.5 (3·[8.5-6]) + 22.59 (30·0.753)
= 68.0
→ normalise(21.9, 12.4, 68.0)
```

→ **Protection 21 · Income 12 · Growth 67**

Solid growth profile; the 22% NVDA position is the main thing lifting Protection above its floor.

Ben **STATED-GOAL** · Retail · \$29.0m · “Moderate, value tilt” · horizon 6.5y

Funding is dual: “Fund \$0.8m/yr income; grow to \$45m by 2033” — both signals fire.

$$\text{Willingness} = 0.5 \cdot 0.50 \text{ (stated)} + 0.5 \cdot [0.5 \cdot 0.82(R) + 0.3 \cdot 0.745(D) + 0.2 \cdot 0.72(C) = 0.778] = \mathbf{0.639}$$

```
requiredReturn = (45/29)^(1/7) - 1 = 6.48%    · incomeNeed = 0.8/29 = 2.76%
PROTECTION = 8 + 3.0 (18%-15 conc) + 10.11 (caution 28·0.361)
= 21.1
INCOME      = 8 + 22.08 (8·2.76 draw) + 6.5 (caution 18·0.361)
= 36.6
GROWTH      = 8 + 32.4 (5·6.48) + 1.5 (3·[6.5-6]) + 19.17 (30·0.639)
= 61.1
→ normalise(21.1, 36.6, 61.1)
```

→ **Protection 18 · Income 31 · Growth 51**

His \$0.8m income goal on a small \$29m book (a demanding 2.8% draw) is what lifts Income to 31 — exactly why his book is “behind” on income at only 6% fixed income.

Morgan **STATED-RISK** · Professional · \$38.0m · “Aggressive — maximum growth” · horizon 17.5y · NO goal

No funding goal on file. 92% risk assets — 24% NVDA, 18% direct BTC, leveraged Nasdaq, venture.

$$\text{Willingness} = 0.5 \cdot 0.95 \text{ (stated)} + 0.5 \cdot [0.5 \cdot 0.92(R) + 0.3 \cdot 0.855(D) + 0.2 \cdot 0.96(C) = 0.908] = \mathbf{0.929}$$

```
no required-return signal → Wv = 45 (lean harder on appetite)
PROTECTION = 8 + 9.0 (24%-15 conc) + 1.99 (caution 28·0.071)
= 19.0
INCOME      = 8 + 1.28 (caution 18·0.071)
= 9.3
GROWTH      = 8 + 34.5 (3·[17.5-6] horizon) + 41.8 (45·0.929)
= 84.3
→ normalise(19.0, 9.3, 84.3)
```

→ **Protection 17 · Income 8 · Growth 75**

No goal number exists, so Growth comes *entirely* from temperament (long horizon + a 0.93 willingness via the no-goal coefficient). The only counterweight is his 24% NVDA lump. Confidence 0.70 — no hard goal to anchor on.

Tejpaul **REVEALED** · Retail · \$60.0m · not formally profiled · horizon 8.5y · NO goal

Nothing stated at all — no risk profile, no targets, no funding goal, no liabilities. The book is the only signal: balanced ~57% equity, ~35% defensive, no concentrated name.

Willingness = revealed only (stated = null) = $0.5 \cdot 0.57(R) + 0.3 \cdot 0.364(D) + 0.2 \cdot 0.36(C)$
= **0.466**

```
no goal, no stated appetite → Wv = 45
PROTECTION = 8 + 14.95 (caution 28·0.534)
= 22.9
INCOME      = 8 + 9.61 (caution 18·0.534)
= 17.6
GROWTH      = 8 + 7.5 (3·[8.5-6] horizon) + 20.97 (45·0.466)
= 36.5
→ normalise(22.9, 17.6, 36.5)
```

→ **Protection 30 · Income 23 · Growth 47**

The purest case: with no form to read, the system reads his balanced book and **independently concludes “moderate”** — Growth 47, with a defensive tilt from the caution term. Confidence 0.55 (lowest tier — confirm with the client).

4 • The whole book at a glance

Client	Source	Protection	Income	Growth	Willingness	Conf.
Morgan	STATED-RISK	17	8	75	0.93	0.70
Fable	STATED-GOAL	16	8	76	0.93	0.90
Prahnav	STATED-GOAL	21	12	67	0.75	0.90
Amar	STATED-GOAL	22	17	61	0.65	0.90
Aurora	STATED-GOAL	27	13	60	0.75	0.90
Ben	STATED-GOAL	18	31	51	0.64	0.90
Jacob	STATED-GOAL	22	29	49	0.51	0.90
Tejpaul	REVEALED	30	23	47	0.47	0.55
Scott	STATED-RISK	39	46	15	0.26	0.70

The spread is the point: from **Morgan/Fable** (~75% Growth) down to **Scott** (15% Growth, 46% Income), every vector is traceable to facts — a required return, a mortgage, a spending draw, a concentrated name, or, where nothing is stated, the shape of the book itself.

Status & next steps. `deriveGoals` is live and exposed as `window.GOALS`, but **additive** — the fit engine and UI still read the legacy five-bucket `goals.target`. Open items: (1) fold 5 buckets → 3 across the app; (2) surface the inferred vector + drivers + an override editor in the Advisor Book; (3) capture a numeric spending field so qualitative “income needs” (e.g. Aurora) are no longer lost. All coefficients above live in `PARAMS` and are tunable.

Brokerage Playground — Goal Derivation · Prepared 18 June 2026 · Source of truth: `goals.js`, `data.js` · All figures generated by running `deriveGoals()` against the live seed book. Methodology: liability-driven / goals-based (Chhabra risk-allocation). Engine is additive — not yet wired into fit scoring.